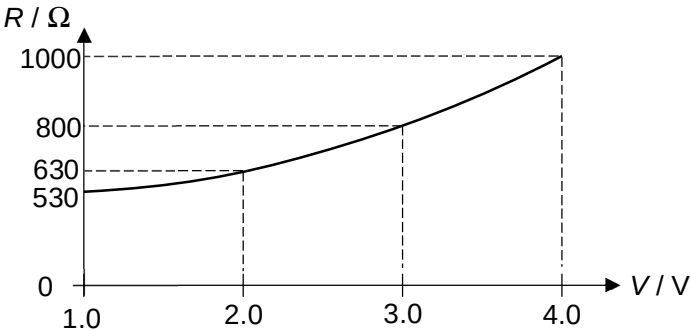


6(a)	When $V = 1.4 \text{ V}$, $I = 2.5 \text{ mA}$ p.d. across variable resistor $= 4.0 - 1.4 = 2.6 \text{ V}$ $\text{resistance} = \frac{2.6}{2.5 \times 10^{-3}}$ $\approx 1040 \, \Omega$ OR $\text{resistance of lamp} = \frac{1.4}{2.5 \times 10^{-3}} = 560 \, \Omega$ $\frac{560}{560 + R_V} \times 4.0 = 1.4$ $R_V \approx 1040 \, \Omega$	
6(b)	 The graph shows the relationship between Resistance (R / Ω) on the vertical axis and Voltage (V / V) on the horizontal axis. The curve starts at $(1.0, 530)$ and increases non-linearly. Key points marked on the curve are $(2.0, 630)$, $(3.0, 800)$, and $(4.0, 1000)$.	
6(c)	When the temperature of the filament increases, the magnitude of the vibration of the lattice ions increases. This causes electrons to collide with the lattice ions more frequently, thus increasing the resistance.	
7(a)	The magnetic flux through any surface area is defined as	